

Lung-control case study: integration reduces structure but does not prove exchangeability

A

Study compatibility map

Study	Samples	Modality	Protocol class	Use in argument
GSE132771	6	scRNA-seq	lineage negative enriched	diagnostic only
GSE159354	5	scRNA-seq	FACS enriched	diagnostic only
GSE171524	7	snRNA-seq	frozen autopsy single nucleus	sensitivity only
GSE173896	6	scRNA-seq	whole tissue scRNA	controlled primary
GSE227691	4	scRNA-seq	whole tissue scRNA	controlled primary

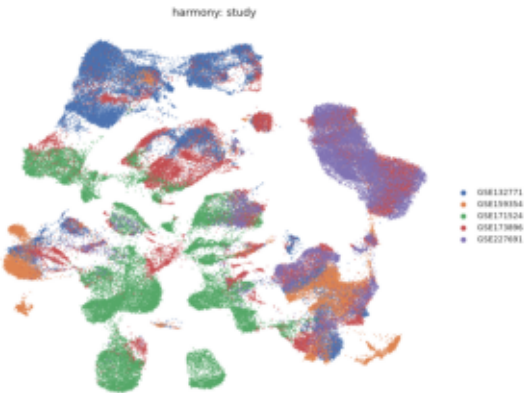
B

Unintegrated



C

Harmony



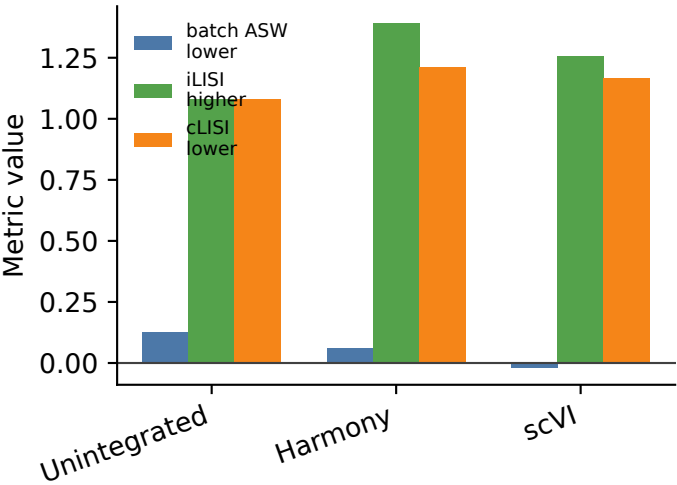
D

scVI



E

Integration diagnostics



F

Donor-level evidence retained after protocol-aware filtering

- 28 public lung control samples from five source studies.
- Controlled whole-tissue scRNA-seq subset: GSE173896 + GSE227691.
- All-study composition was study-associated; controlled subset divergence was lower but non-zero.
- Pseudobulk state and differential-expression screens remained study-associated in several cell types.
- Interpretation: residual study-conditioned heterogeneity, not a single biological mechanism.